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Tesla-SolarCity

We're going to look back on gasoline cars like we look back at steam engines. It was a phase, it was a bit weird. We're going to look back on fossil fuel power generation the same way: it was a weird phase. We want to get out of that weird phase as soon as we can … so, yeah, we're trying to have the non-weird future get here as fast as possible.

— Tesla CEO Elon Musk (June 22, 2016)[1]

Joan Bannister, an equity analyst at the Katalyst Group, dialed into a conference call hosted by Tesla Motors, Inc. ("Tesla") and SolarCity Corporation ("SolarCity") on August 1, 2016 to discuss the merger agreement they had announced the previous day. On the call, executives from Tesla and SolarCity reiterated their strategic rationale for the business combination and described the details of the merger agreement. Specifically, Tesla would acquire SolarCity in an all-stock transaction in which SolarCity shareholders would receive 0.110 shares of Tesla stock for each of their SolarCity shares. The transaction would be subject to the approval of a majority of "disinterested" shareholders (e.g., any executive or board member who had a conflict of interest) of both Tesla and SolarCity. If approved, the deal was expected to close in the fourth quarter of 2016, after which current Tesla shareholders would own approximately 93.1% of the combined entity and SolarCity shareholders would own approximately 6.9%.

The Tesla-SolarCity merger announcement was not completely unexpected. Six weeks earlier, on June 21, 2016, Tesla had made an offer to acquire SolarCity at an exchange ratio between 0.122 to 0.131 shares of Tesla stock (subject to due diligence) for each share of SolarCity, which Tesla represented as a premium of between 21% to 30% over SolarCity's previous closing price.[2] In making its offer, Tesla argued that the combination of Tesla and SolarCity would provide significant benefits to the shareholders, customers, and employees of both companies. Tesla elaborated:

> We would be the world's only vertically integrated energy company offering end-to-end clean energy products to our customers. This would start with the car that you drive and the energy that you use to charge it, and would extend to how everything else in your home or business is powered… Because of the shared ideals of the companies and our customers, those who are interested in buying Tesla vehicles or Powerwalls are naturally interested in going solar, and the reverse is true as well.[3]

Senior Lecturer E. Scott Mayfield and Professor Emil N. Siriwardane prepared this case. This case was developed from published sources. Funding for the development of this case was provided by Harvard Business School and not by the company. Joan Bannister and Katalyst Group are fictional. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Finally, Tesla explained that each company would be able to build on the core competencies of the other and that, because both companies were "driven by a mission of sustainability, innovation, and overcoming challenges," their cultures would be a great fit.[4]

Although Tesla had articulated numerous reasons for the proposed merger, investors and analysts were more circumspect. Many analysts questioned the rationale for the merger and expressed concerns that the merger could distract management from important near-term Tesla production goals.[5] Some analysts went so far as to suggest that the offer was, in essence, a "bailout" of SolarCity, which was facing potential liquidity issues.[6] On the day following the announcement of the initial June merger offer, Tesla's stock price fell 10.5%, from $219.61 per share to $196.66 per share, while SolarCity's stock price rose 3.3% from $21.19 per share to $21.88 per share. During the subsequent six weeks, Tesla's stock recovered from its initial loss. On July 29, 2016, Tesla's stock price closed at $234.79 per share and SolarCity's stock price closed at $26.70 per share, increasing 6.9% and 26.0%, respectively, from their values before the announcement of the merger offer. (**Exhibit 1** shows the stock prices for Tesla and SolarCity over the period from June 1 through July 29, 2016.)

Bannister found the recovery in Tesla's stock price somewhat comforting, yet the market's harsh initial reaction still concerned her. (See **Table A**.) As Tesla's conference call came to a close, Bannister reflected on what she had heard and wondered how investors would react to news of the merger agreement when the markets opened later that morning. More importantly, she wondered what she should recommend to her clients who held various positions in Tesla and SolarCity. She looked at her watch; it was 9:10am and her phone was ringing.

Table A Tesla and SolarCity Stock Price Reaction (Percent change from June 21, 2016)

	Tesla		SolarCity		Nasdaq
	$	% Chg	$	% Chg	% Chg
June 21, 2016	219.61		21.19		
June 22, 2016	196.66	-10.5%	21.88	3.3%	-0.2%
July 29, 2016	234.79	6.9%	26.70	26.0%	6.6%

Source: CRSP and Bloomberg.

Tesla Motors, Inc.

Headquartered in Palo Alto, California, Tesla[a] was founded in 2003 with the near-term goal of building sports cars powered by lithium-ion batteries. Elon Musk had been a major investor in Tesla's Series A financing round and became its Chairman in 2004. In 2008, Musk assumed the role of CEO and, as of July 2016, owned over 20% of Tesla's equity. In 2015, Tesla reported revenues of $4.0 billion and, as of June 30, 2016, had a market capitalization of over $31 billion, ranking it among the largest U.S. automakers based on market capitalization. (**Exhibits 2, 3, and 4** show Tesla's recent financial statements.) In addition to equity, Tesla's operations were financed with a short-term credit facility and

[a] Martin Eberhard, one of Tesla's original founders, named the company after the Serbian inventor Nikola Tesla (1856-1943), who patented the type of alternating current (AC) induction motor used to power the Tesla roadster.

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three convertible notes. (**Exhibit 5** provides market values for Tesla's outstanding debt and **Exhibits 6a** and **6b** contain information on Tesla's convertible notes.)

Tesla's primary production facility was located in nearby Fremont, California.[b] Since the introduction of its Model S sedan in 2012, Tesla had grown quickly and its two main electric cars, the Model S sedan and the Model X sports utility vehicle, were at the top of the luxury market in terms of critical reviews and performance.[7] In 2015, Tesla delivered 50,000 vehicles, bringing total deliveries since 2012 to 107,000.[c] Tesla planned to introduce a lower-cost, "mass-market" model (the "Model 3") in 2017 and the firm anticipated the new Model 3 would be in large demand.[d] With the introduction of the Model 3, Tesla expected to increase its annual production of new vehicles to 500,000 units by 2018. Analysts viewed the introduction of the Model 3 as a pivotal event for Tesla, with one analyst going as far as to call it Tesla's "iPhone moment."[8]

To support the production of its vehicles, Tesla broke ground on its "Gigafactory" in 2014. According to projections, the Gigafactory would make Tesla the world's largest producer of lithium-ion batteries and provide sufficient capacity to supply all of the batteries for its automobiles. In addition, the Gigafactory would also manufacture the Tesla "Powerwall", a rechargeable battery system designed for home energy storage. Utilizing the same lithium-ion battery technology that powered Tesla vehicles, the Powerwall took energy from solar sources or from the electric grid and allowed residents to power their homes at night or in the event of an outage, all from the touch of a mobile-phone based application. Importantly, the Powerwall provided the link to Musk's broader vision of a "sustainable energy" future.

Musk had a long-term vision of providing consumers with a viable alternative to fossil fuels. In his 2006 "Secret Tesla Motors Master Plan", Musk wrote: "the overarching purpose of Tesla Motors … is to help expedite the move from a mine-and-burn hydrocarbon economy toward a solar electric economy." Summarizing his plan, Musk concluded:

> So, in short, the master plan is: Build sports car
> Use that money to build an affordable car
> Use *that* money to build an even more affordable car
> While doing above, also provide zero emission electric power generation options[9]

To this end, Musk encouraged his cousins, Peter and Lyndon Rive, to start a company devoted to solar power generation. The Rives founded SolarCity in 2006 with Musk as Chairman. As of July 2016, Musk owned over 20% of SolarCity's equity.[10] Discussing his master plan, Musk explained that there were three important components to a sustainable energy future: generation, storage, and consumption. Tesla's Powerwall allowed consumers to efficiently store the energy they harvested from the sun, and the fleet of Tesla's electric vehicles provided a more environmentally friendly way to consume this energy. SolarCity was the last piece of the puzzle in terms of providing low-emission

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[b] Tesla purchased the Freemont facility from Toyota's New United Motor Manufacturing Inc. (NUMMI), which operated the plant from 1984 to 2009. From 1962 to 1982, the facility had been operated exclusively by General Motors, Inc. Output at the facility peaked in 2006 at nearly 430,000 vehicles. Source: https://www.tesla.com/factory and http://wardsauto.com/news-analysis/toyota-s-decision-abandon-nummi-closes-book-25-year-experiment.

[c] In comparison, global sales for General Motors totaled just under 10 million vehicles for 2015. Source: http://media.gm.com/media/us/en/gm/home.detail.html/content/Pages/news/us/en/2016/Jan/0121-global-sales.html.

[d] As of May 4, 2016, 325,000 potential customers had paid deposits of $1000 each to reserve the Model 3, despite the fact that there had been no advertising and there was no specific date for delivery. Source: Tesla Q1' 16 Update Letter, http://ir.tesla.com/downloads.cfm?view=all

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power generation. Musk stated: "This is the integrated future. You've got an electric car, a Powerwall, and a Solar Roof. It's pretty straightforward, really."[11]

SolarCity Corporation

SolarCity designed and manufactured solar panels for residential and commercial use and was founded on the idea that solar panels would eventually replace utility-generated power as the predominant source of energy for households, businesses, and governments. Since its inception, SolarCity had installed more than 230,000 solar energy systems for customers in 27 different states.[12] In 2015, SolarCity reported revenues of $400 million, up 57% from the previous year and, as of June 21, 2016 had a market capitalization of $2.2 billion. (**Exhibits 7, 8,** and **9** show SolarCity's recent financial statements.)

Since 2008, the "SolarLease" had been a key element of SolarCity's sales strategy. When a customer chose this financing option, SolarCity would install solar panels on the customer's home at no upfront cost and, in return, the customer would enter into 20-year lease agreement with SolarCity.[e] These transactions typically occurred through subsidiaries that were owned by SolarCity and other minority equity investors. Because SolarCity incurred the upfront installation cost but realized the lease payments over 20 years, its rapid growth had caused its cash flow from operations to be negative. For example, during the first six months of 2016, SolarCity's cash flow from operations was negative $389 million. For this reason, SolarCity was highly dependent on external funding sources, specifically debt capital.

To finance its growth, SolarCity had relied on four types of debt capital: (1) short-term bank loans that were secured primarily by its working capital, (2) non-recourse long-term bonds that were secured by customer lease payments on its solar installations,[f] (3) "solar bonds" that were offered directly to retail consumers interested in making "green" investments, and (4) convertible notes. SolarCity's solar bonds and convertible notes were both unsecured and structurally subordinate (i.e., lower priority in bankruptcy) to SolarCity's short-term bank loans and long-term bonds. As of June 30, 2016, SolarCity had total debt outstanding of $3.2 billion, of which solar bonds and convertible notes accounted for $218 million and $909 million face value, respectively.[g] (**Exhibit 10** provides market values for SolarCity's outstanding debt.)

In December 2015, an adverse regulatory change in Nevada caused SolarCity to revise its guidance for the first quarter of 2016 downward. Specifically, the Nevada state utilities commission eliminated net metering[h] for new and existing retail solar customers, forcing SolarCity to shut down its operations in the state.[13] Nevada had been expected to account for 6.5% of SolarCity's installed megawatts in 2016 and the ruling caused analysts to question whether SolarCity would be able to achieve its 2016 projections.[14] This development, combined with declining oil and gas prices that made solar relatively less attractive, led some analysts to express concern about SolarCity's ability to meet its growth forecasts and raise debt at reasonable terms.[15] To make matters worse, in March 2016, a rumor regarding the potential bankruptcy

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[e] SolarCity would retain ownership of the panels and receive any tax incentives associated with their installation.

[f] SolarCity's non-recourse debt consisted primarily of long-term bonds issued by its subsidiaries. Holders of these non-recourse bonds had claims against the subsidiary that issued the debt and not the parent SolarCity.

[g] Musk's private space exploration firm, SpaceX, owned $90 million and $75 million of solar bonds due in March and June of 2017, respectively. Elon Musk and Lyndon Rive owned $13 million and $3 million of SolarCity's convertible notes due in 2020, respectively.

[h] Net metering allowed the owner of a solar energy system to receive credits for the electricity she supplied to the power grid. Specifically, if a residential customer generated more electricity than she used, her electric meter would "run backward," such that she was only billed for her "net" energy usage.

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of a major competitor caused the stock prices of several solar firms, including SolarCity, to fall significantly. By June 21, 2016, SolarCity's stock price was down 52% from its value on December 31, 2015 and its convertible notes were trading at prices averaging 62% of par value, implying a yield of approximately 25% on the straight-bond component. (**Exhibits 11a** and **11b** provide information on SolarCity's convertible notes.) In comparison, Tesla's convertible notes implied a straight-bond yield of 7.4% (see **Exhibit 6b**) and the yield on CCC rated corporate bonds was 14.4% at the time.[i]

The Merger Opportunity

Musk saw the Tesla-SolarCity merger as an important part of his goal to accelerate the advent of sustainable energy "so that we can imagine far into the future and life is still good."[16] To make this goal a reality, Musk planned to create an "integrated and beautiful solar-roof-with-battery product that just works… one ordering experience, one installation, one service contract, one phone app."[17] Toward this end, Musk explained why combining Tesla and SolarCity was crucial:

> We can't do this well if Tesla and SolarCity are different companies, which is why we need to combine and break down the barriers inherent to being separate companies. That they are separate at all, despite similar origins and pursuit of the same overarching goal of sustainable energy, is largely an accident of history. Now that Tesla is ready to scale Powerwall and SolarCity is ready to provide highly differentiated solar, the time has come to bring them together.[18]

In addition to accelerating the transition to sustainable energy, Tesla argued that the combination of solar and storage technologies would create unique benefits for investors. By bringing Tesla's Powerwall product and international retail distribution network together with SolarCity's "best-in-class" installation costs and extensive retail sales network, the combination would extend the reach of both firms and generate $150 million in annual direct cost synergies as a result of sales and marketing efficiencies and corporate and overhead savings.[19] Based on discount rates used by advisors to SolarCity, the projected cost savings would have a capitalized value of about $975 million.[j]

Subsequent to the announcement of the offer in June, Tesla and SolarCity each formed Special Committees comprised of independent board members, as defined in the merger agreement, to provide recommendations to their respective shareholders regarding the merger offer. (See **Exhibit 12** for the composition of both firm's boards.) Each committee hired an outside advisor to conduct valuation analyses of both Tesla and SolarCity and to opine on the reasonableness of the proposed offer. Following this due diligence process, Tesla and SolarCity agreed that Tesla would pay 0.110 shares of Tesla stock for each share of SolarCity stock, less than the originally offered range of 0.122 to 0.131. Similarly, the conversion ratio for SolarCity's convertible debt would be adjusted in like fashion, i.e., the conversion ratio would be multiplied by 0.110.[k] All other debt of the two companies was expected to remain outstanding with unchanged terms through the completion of the merger.

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[i] Corporate bond yield data from Capital IQ.

[j] Assumes growth in projected cost savings of 2.5% per year, a tax rate of 35%, and a discount rate of 12.5%. As part of the due diligence process, the financial advisor (Lazard Frères & Co. LLC) to the SolarCity Special Committee estimated a range for the weighted-average cost of capital for Tesla of 12% to 13%. Source: Tesla Form S-4 filed on September 19, 2016, page 98.

[k] SolarCity's convertible debt contained "Fundamental Change" clauses that, in some circumstances, could require a further adjustment to the conversion ratio or repayment of the instrument. However, because the transaction was structured as a merger with Tesla stock as consideration, it did not trigger any of these conditions.

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In addition to specifying the exchange ratio for the transaction, the Tesla-SolarCity agreement included two important features. As part of the agreement, SolarCity would have a 45-day "go shop" period, during which it would be allowed to solicit proposals from other potential merger partners. And, finally, the deal would need to be approved by a majority of the "disinterested" shareholders of both Tesla and SolarCity, as defined in the merger agreement.[20] In essence, any executive or board member who had a conflict of interest, including Musk, would not be included in the shareholder vote. Each company's special committee recommended that shareholders vote to approve the merger.

Analyst Reaction

Analyst reaction to the proposed merger was mixed. Although many analysts acknowledged the potential for synergies between the two firms, they also expressed concerns about the stresses that the merger could impose. For example, analysts at Piper Jaffray wrote:

> We think this potential deal makes a lot of strategic sense. By attempting to produce a full suite of consumer products that produce, store, and consume energy, TSLA is demonstrating once again how ambitious its long-term strategy really is…. However, with the Model 3, TSLA is already subjecting itself to one of the most ambitious (and probably one of the least achievable) production ramps in automotive history. Integrating SCTY wouldn't make this process any easier. [21]

Similarly, analysts at Deutsche Bank wrote:

> Overall, while we see potential for synergies, and we acknowledge that the acquisition of Solar City would only increase Tesla's outstanding shares by 8-9%, we believe that Tesla Investors will generally view this development as a negative, as it could produce a distraction while management would benefit more by focusing on their impressive growth plans (i.e. growing from 50k to 500k to 1,000k units). [22]

Other analysts were more critical, however, and expressed concerns about possible conflicts of interest, the magnitude of potential synergies, and whether a merger was required to realize the synergies:

> UBS: There are [SCTY] 8 board members. Musk is conflicted (>20% ownership in both companies). It's also known that Lyndon & Peter Rive are first cousins of Musk. In addition, Straubel is TSLA CTO. This leaves only 4 independent members. Gracias is conflicted being on both boards, Fisher's firm (Draper Fisher Jurvetson) is also on both boards, and Pfund's firm was an early investor in TSLA. Arguably this leaves Donald Kendall Jr. as the lone member on SCTY's board without any apparent Tesla relationship… [23]
>
> JP Morgan: We believe investors may initially struggle to identify material synergies from combining the two businesses, due to the widely different customer sets, sales channels, and technologies. We see relatively little overlap of core competencies across vehicle manufacturing and solar panel installations. [24]
>
> Credit Suisse: Why not pursue a less capital-intensive matter, at least in the near term, and proceed with more of a marketing or sales relationship between the entities with the cross-sell opportunities and installation efficiencies of the integrated systems? [25]

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Conclusion

Bannister had many questions to answer. Was the merger with SolarCity an opportunity, as Tesla had portrayed? Or was it a distraction and a bailout of SolarCity, as some analysts had suggested? Was Musk's vision of a sustainable energy future soon to be a reality? Or would the desire for profits start to weigh on investors' patience? The stock market had initially reacted negatively to the merger but had subsequently seemed more receptive to the idea. Bannister was also aware, however, that SolarCity bondholders had reacted positively to the deal. She noted that, in response to the merger offer on June 21, 2016, the price of SolarCity convertible debt had jumped almost 20% and was up about 25% by July 29, 2016.[1] (**Exhibit 13** graphs the price of SolarCity's convertible debt over the period from June 1, 2016 to July 29, 2016.) Bannister's clients had varied interests: some had investments in Tesla; some had investments in SolarCity; some held equity positions; and others held debt positions. Needing to understand these different perspectives, Bannister had a lot to think about and little time… as her phone continued to ring.

[1] After accounting for the value of the option to convert the notes to equity, which decreased in value, the implied increase in the value of the straight-bond component of SolarCity's 1.625% convertible note due 2019 was 33.4% on the day after the merger announcement and was 42.1% by July 29, 2016. The 1.625% convertible note was SolarCity's largest outstanding convertible note.

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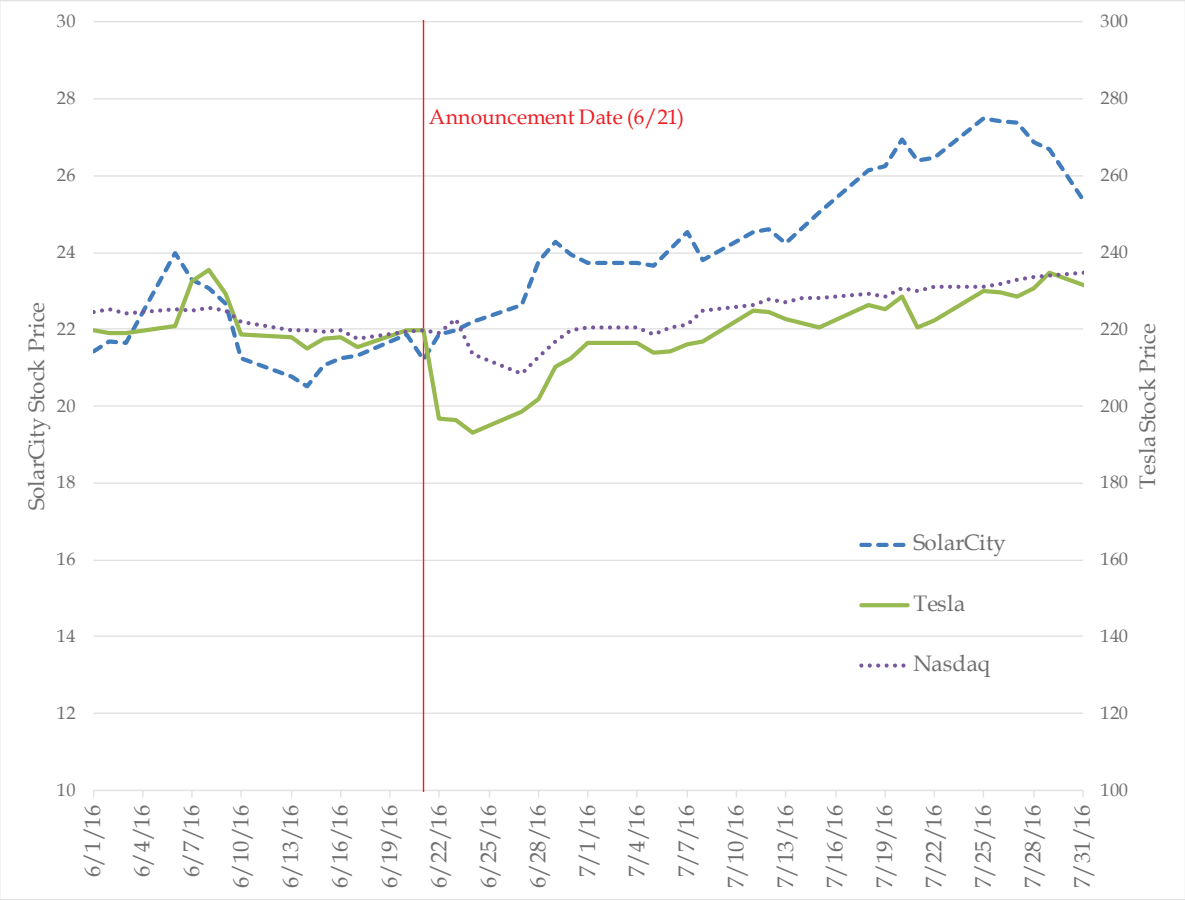
Exhibit 1 Tesla and SolarCity Stock Prices: June 1, 2016 – July 31, 2016 ($)



Sources: Center for Research in Security Prices.

Note: Nasdaq indexed to Tesla stock price on 6/21/2016.

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Exhibit 2 Tesla Income Statements, 2014-2016 ($ millions)

	12 Mos. ending Dec. 31		6 Mos. Ending
	2014	2015	June 30, 2016
Operating revenue	**3,198**	**4,046**	**2,417**
Cost of goods sold	2,317	3,123	1,890
Gross profit	882	924	527
SG&A	604	922	639
R&D	465	718	374
Operating income	**(187)**	**(717)**	**(486)**
Interest expense, net	100	117	84
Other income (expense)	2	(42)	2
Income tax	9	13	7
Net income to common	**(294)**	**(889)**	**(575)**

Source: Capital IQ.

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Exhibit 3 Tesla Balance Sheets, 2014-2016 ($ millions, unless stated otherwise)

	December 31		June 30
	2014	2015	2016
Assets			
Cash and equivalents	1,906	1,197	3,246
Accounts receivable	227	169	179
Inventory	954	1,278	1,610
Other current assets	94	138	169
Total current assets	**3,180**	**2,782**	**5,204**
Net PP&E	2,596	5,195	6,527
Other LT Assets	55	91	138
Total assets	**5,831**	**8,068**	**11,869**
Liabilities and stock holders' equity			
Accounts payable	778	916	1,115
Accrued expenses	123	227	375
Current maturities of LT debt	669	675	664
Other current liabilities	595	1,040	1,649
Total current liabilities	**2,165**	**2,858**	**3,804**
Long-term debt	1,819	2,021	2,620
Other liabilities	935	2,105	2,925
Total Liabilities	**4,919**	**6,984**	**9,349**
Common equity	912	1,084	2,520
Total liabilities & equity	**5,831**	**8,068**	**11,869**
Other Information			
Credit rating	B-	B-	B-
Shares outstanding (Millions)	126	131	148
Stock price per share ($)	222.41	240.01	212.28

Source: Capital IQ.

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Exhibit 4 Tesla Cash Flow Statements, 2014-2016 ($ millions, unless stated otherwise)

	12 Mos. ending Dec. 31		6 Mos. Ending
	2014	2015	June 30, 2016
Operating Activities			
Net Income	(294)	(889)	(575)
Depreciation & Amort., Total	232	423	340
Other Amortization	70	78	42
(Gain) Loss From Sale Of Assets	14	38	12
Stock-Based Compensation	156	198	157
Other Operating Activities	21	121	38
Change in Acc. Receivable	(184)	46	(1)
Change In Inventories	(1,050)	(369)	(1,218)
Change in Acc. Payable	415	263	213
Change in Unearned Rev.	210	322	165
Change in Other Net Operating Assets	353	(756)	730
Cash from Operations	**(57)**	**(525)**	**(99)**
Investing Activities			
Capital Expenditure	(970)	(1,635)	(512)
Cash Acquisitions	-	(12)	-
Invest. in Marketable & Equity Securt.	(17)	-	17
Other Investing Activities	(4)	(26)	(59)
Cash from Investing	**(990)**	**(1,674)**	**(554)**
Financing Activities			
Total Debt Issued	2,303	888	1,493
Total Debt Repaid	(11)	(204)	(597)
Issuance of Common Stock	100	857	1,812
Other Financing Activities	(249)	(17)	(16)
Cash from Financing	**2,143**	**1,524**	**2,692**
Foreign Exchange Rate Adj.	(36)	(34)	10
Net Change in Cash	**1,060**	**(709)**	**2,049**

Source: Capital IQ.

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Exhibit 5 Composition and Market Value of Tesla Outstanding Debt ($ millions)

	Par Value on	Market Value on		
	June 30, 2016 [2]	June 21, 2016	June 22, 2016	July 29, 2016
Revolving Credit Facility [1]	678	678	678	678
Convertible Notes	2,946	3,264	3,019	3,271
Total Debt Instruments	3,624	3,942	3,697	3,949

Source: Capital IQ, Datastream, and casewriter estimates.

Notes: [1] Analysis assumes that the market value of the revolving credit facility equals par value because it was secured.

[2] Par value of total debt instruments does not equal the sum of current maturities of long-term debt and long-term debt outstanding reported in **Exhibit 3**. In its financial statements, Tesla reported the option component of its convertible debt as an equity item.

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Exhibit 6a Information for Tesla Convertible Notes ($ millions, unless stated otherwise)

Issue Date	Maturity Date	Coupon (%)	Par Value	Conversion Ratio	Price ($) per $1000 of par on 6/21/2016	6/22/2016	7/29/2016
5/22/2013	6/1/2018	1.500%	646	8.0306	1,794.4	1,620.3	1,876.3
3/5/2014	3/1/2019	0.250%	920	2.7788	933.7	889.9	922.9
3/5/2014	3/1/2021	1.250%	1,380	2.7788	902.7	835.5	876.6

Source: Datastream.

Exhibit 6b Value Decomposition of Tesla's 1.25% Convertible Note Due 2021

	6/21/2016	6/22/2016	7/29/2016
Tesla Stock Price (TSLA)	219.61	196.66	234.79
Strike Price	359.87	359.87	359.87
Maturity (years)	4.7	4.7	4.6
TSLA Impiled Volatility (3-Year)	43%	47%	45%
Risk Free Rate	1.22%	1.22%	1.03%
TSLA Convertible Note Price	**902.7**	**835.5**	**876.6**
Black-Scholes Option Component	138.4	126.4	164.9
Implied Straight-Bond Component	764.3	709.1	711.8
Implied Straight-Bond Yields	**7.4%**	**9.1%**	**9.2%**

Source: Bloomberg, Datastream, the Federal Reserve, and casewriter estimates.

Note: Tesla's 1.25% convertible note due 2021 was its largest outstanding and longest maturity convertible note.

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Exhibit 7 SolarCity Income Statement, 2014-2016 ($ millions)

	12 Mos. ending Dec. 31		6 Mos. Ending
	2014	2015	June 30, 2016
Revenues	**255**	**400**	**308**
Cost of goods sold	176	281	207
Gross profit	79	119	101
SG&A	395	702	416
R&D	19	65	28
Restructuring, other	-	-	65
Operating income	**(335)**	**(648)**	**(408)**
Interest expense, net	56	104	70
Other income (expense)	(11)	(14)	(56)
Income tax	(27)	3	(0)
Net income to company	(375)	(769)	(533)
Minority interest in earnings	319	710	453
Net income to common	(56)	(58)	(80)

Source: Capital IQ.

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Exhibit 8 SolarCity's Balance Sheets, 2014-2016 ($ millions, unless stated otherwise)

	December 31		June 30
	2014	2015	2016
Assets			
Cash and equivalents	504	383	146
Short-term investments	138	11	-
Accounts receivable	53	46	79
Inventory	217	343	230
Other current assets	85	120	156
Total current assets	**998**	**902**	**611**
Net PP&E	102	547	909
Other long-term assets [1]	3,452	5,838	6,703
Total assets	**4,551**	**7,287**	**8,224**
Liabilities and stock holders' equity			
Accounts payable	238	365	214
Accrued expenses	95	144	155
Current maturities of LT debt	26	372	467
Other current liabilities	208	312	265
Total current liabilities	**567**	**1,193**	**1,101**
Long-term debt	1,356	2,344	2,762
Other liabilities	1,286	2,015	2,490
Total Liabilities	**3,209**	**5,553**	**6,353**
Minority interest	597	856	981
Common equity	746	879	891
Total liabilities & equity	**4,551**	**7,287**	**8,224**
Other Information			
Credit rating	NR	NR	NR
Shares outstanding (Millions)	97	98	100
Stock price per share ($)	53.48	51.02	23.93

Source: Capital IQ.

Note: [1] Other long-term assets consist primarily of solar energy systems leased to customers.

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Exhibit 9 SolarCity Cash Flow Statements, 2014-2016 ($ millions, unless stated otherwise)

	12 Mos. ending Dec. 31		6 Mos. Ending
	2014	2015	June 30, 2016
Operating Activities			
Net Income	(56)	(58)	(80)
Depreciation & Amort., Total	98	167	129
(Gain) Loss From Sale Of Assets	1	4	1
(Gain) Loss On Sale Of Invest.	-	12	51
Asset Writedown & Restructuring Costs	-	-	15
Stock-Based Compensation	66	86	44
Tax Benefit from Stock Options	-	(63)	6
Other Operating Activities	(377)	(742)	(476)
Change in Acc. Receivable	1	(11)	(30)
Change In Inventories	(97)	(125)	114
Change in Acc. Payable	112	125	(150)
Change in Unearned Rev.	138	12	46
Change in Other Net Operating Assets	(104)	(196)	(59)
Cash from Operations	**(218)**	**(790)**	**(389)**
Investing Activities			
Capital Expenditure	(1,186)	(1,842)	(902)
Cash Acquisitions	2	(10)	(14)
Invest. in Marketable & Equity Securt.	(161)	125	11
Other Investing Activities	-	-	(9)
Cash from Investing	**(1,345)**	**(1,727)**	**(913)**
Financing Activities			
Total Debt Issued	1,234	1,581	1,089
Total Debt Repaid	(358)	(245)	(600)
Issuance of Common Stock	20	12	2
Other Financing Activities	594	1,047	574
Cash from Financing	**1,490**	**2,395**	**1,065**
Net Change in Cash	**(73)**	**(122)**	**(237)**

Source: Capital IQ.

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Exhibit 10 Composition and Market Value of SolarCity Outstanding Debt ($ millions)

	Par Value on	Market Value on		
	June 30, 2016 [3]	June 21, 2016	June 22, 2016	July 29, 2016
Short-term Bank Loans [1]	383	383	383	383
Non-recourse Debt [1]	1,736	1,736	1,736	1,736
Solar Bonds [2]	218	175	194	200
Convertible Notes	909	563	674	703
Total Debt Instruments	3,246	2,857	2,987	3,022

Source: Capital IQ, Bloomberg, Datastream, and casewriter estimates.

Notes: [1] Analysis assumes that the market value of the short-term bank loans and non-recourse debt equals par value because they were secured and structurally senior to the solar bonds and convertible notes.

[2] Because of data limitations, the market value of SolarCity's solar bonds is estimated using the yield implied by the straight-bond component of SolarCity's 1.625% convertible note due 2019.

[3] Par value of total debt instruments does not equal the sum of current maturities of long-term debt and long-term debt outstanding reported in **Exhibit 8**. In its financial statements, SolarCity reported the option component of its convertible debt as an equity item.

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Exhibit 11a Information for SolarCity Convertible Notes ($ millions, unless stated otherwise)

Issue Date	Maturity Date	Coupon (%)	Par Value	Conversion Ratio	Price ($) per $1000 of par on 6/21/2016	6/22/2016	7/29/2016
10/21/2013	11/1/2018	2.750%	230	16.2165	677.8	786.4	814.0
9/30/2014	11/1/2019	1.625%	566	11.9720	576.1	709.5	739.6
12/7/2015	12/1/2020	0.000%	113	30.3030	715.8	809.8	855.9

Source: Bloomberg and casewriter estimates.

Note: Because of data limitations, the market value of SolarCity's 0% coupon convertible note is estimated using information from SolarCity's 1.625% convertible note due 2019.

Exhibit 11b Value Decomposition of SolarCity's 1.625% Convertible Note Due 2019

	6/21/2016	6/22/2016	7/29/2016
Solar City Stock Price (SCTY)	21.19	21.88	26.70
Strike Price	83.53	83.53	83.53
Maturity (years)	3.4	3.4	3.3
SCTY Impiled Volatility (3-Year)	87%	66%	50%
Risk Free Rate	0.89%	0.89%	0.76%
SCTY Convertible Note Price	**576.1**	**709.5**	**739.6**
Black-Scholes Option Component	73.0	38.4	24.8
Implied Straight-Bond Component	503.1	671.1	714.8
Implied Straight-Bond Yields	**24.8%**	**14.8%**	**13.1%**

Source: Bloomberg, the Federal Reserve, and casewriter estimates.

Note: SolarCity's 1.625% convertible note due 2019 was its largest outstanding and longest maturity convertible note for which pricing data were available.

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Exhibit 12 Tesla and SolarCity Board Members

Name	Title	Tesla Board	SolarCity Board
Brad W. Buss	Former CFO SCTY	X	
Robyn M. Denholm	CFO/COO of Juniper Networks	X	
Ira Ehrenpreis	General Partner with Technology Partners	X	
John H.N. Fisher	Managing Director, Draper Fisher Jurvetson		X
Antonio J. Gracias	CEO, Chairman of Valor	X	X
Steve Jurvetson	Managing Director, Draper Fisher Jurvetson	X	
Donald R. Kendall Jr	Managing Director & CEO of Kenmont		X
Elon Musk	TSLA Chairman and CEO; SCTY Chairman	X	X
Kimbal Musk	CEO of Medium	X	
Nancy E. Pfund	Founder/Managing Partner of DBL Investors		X
Lyndon Rive	SCTY CEO and co-founder		X
Peter Rive	SCTY CTO and co-founder		X
JB Straubel	CTO TSLA		X

Source: UBS Global Research; accessed at https://neo.ubs.com/shared/d109GOl9qV/.

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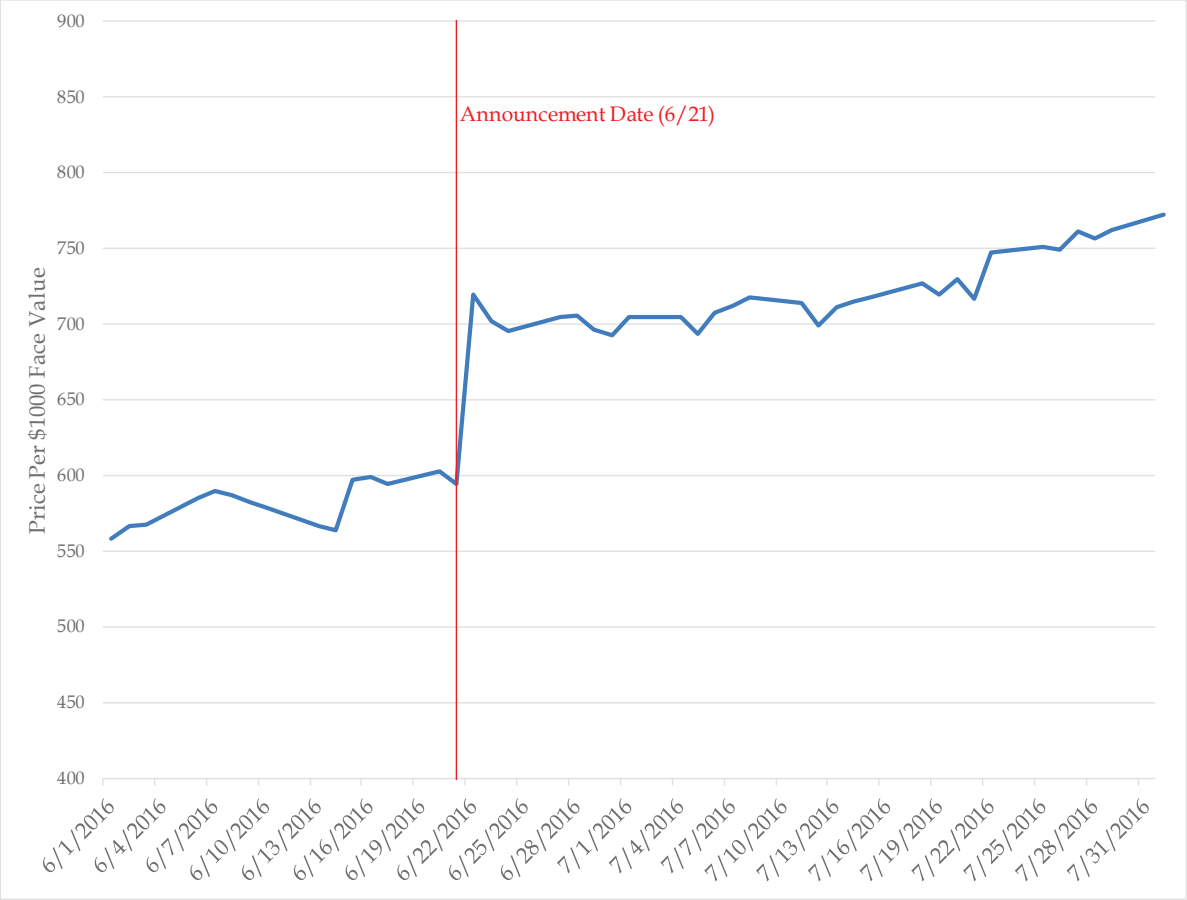
Exhibit 13 Average SolarCity Convertible Debt Price: June 1, 2016 – July 31, 2016 ($)



Source: Bloomberg and casewriter estimates.

Note: The reported average price is the notional-weighted average over SolarCity convertible notes with available pricing. The notional-weighted average maturity of all notes outstanding was approximately 3.25 years on June 21, 2016.

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Endnotes

[1] Tesla conference call, June 22, 1016

[2] Alexander Potter and Winnie Dong, "Buying SCTY Is Neat Idea, but TSLA May Be Asking Too Much of Itself," Piper Jaffray, June 21, 2016, via ThompsonOne, accessed August 2017.

[3] https://www.tesla.com/blog/tesla-makes-offer-to-acquire-solarcity

[4] https://www.tesla.com/blog/tesla-makes-offer-to-acquire-solarcity

[5] Ryan Brinkman, Samik Chatterjee, and Elad Hillman, "Proposed SolarCity Acquisition Adds to Increasingly Binary and Difficult-to-Value Nature of Tesla Story," J.P. Morgan, June 22, 2016, via ThompsonOne, accessed August 2017.

[6] Tesla Motors / SolarCity Conference Call June 22, 2016. https://forums.tesla.com/forum/forums/tesla-motors-solarcity-conference-call-june-22-2016

[7] See https://www.edmunds.com/tesla/model-s/2017/review/. In addition, see https://www.edmunds.com/tesla/model-x/2017/review/

[8] Rod Lache, Mike Levin, Patrick Nolan, and Kirill Babikov, "Tesla's iPhone Moment", Deutsche Bank, March 30, 2016, via Thompson One, accessed March 2018.

[9] https://www.tesla.com/blog/secret-tesla-motors-master-plan-just-between-you-and-me.

[10] https://www.ft.com/content/e8272396-3839-11e6-a780-b48ed7b6126f

[11] https://www.fastcompany.com/40422076/the-real-story-behind-elon-musks-2-6-billion-acquisition-of-solarcity-and-what-it-means-for-teslas-future-not-to-mention-the-planets

[12] https://www.sec.gov/Archives/edgar/data/1408356/000156459016012549/scty-10k_20151231.htm#Item_1

[13] https://www.greentechmedia.com/articles/read/nevada-regulators-eliminate-retail-rate-net-metering-for-new-and-existing-s#gs.1NJoG=k

[14] Steven Byrd, Devin McDermott, David Arcaro, Jim Kobus, and Arman Tabatabai, "Examining the Three Biggest Risks to Rooftop Solar," Morgan Stanley, March 1, 2016, via Thompson One, accessed August 2017.

[15] Steven Byrd, Devin McDermott, David Arcaro, Jim Kobus, and Arman Tabatabai, "Examining the Three Biggest Risks to Rooftop Solar," Morgan Stanley, March 1, 2016, via Thompson One, accessed August 2017.

[16] https://www.tesla.com/blog/master-plan-part-deux

[17] https://www.tesla.com/blog/master-plan-part-deux

[18] https://www.tesla.com/blog/master-plan-part-deux

[19] http://files.shareholder.com/downloads/ABEA-4CW8X0/6048129711x0x902286/19938ABD-D64F-4DEA-A3DB-9116D79CCA5E/Investor_Presentation_Final.pdf

[20] Source: http://files.shareholder.com/downloads/ABEA-4CW8X0/2128854590x0xS1193125-16-665620/1318605/filing.pdf

[21]Alexander Potter and Winnie Dong, "Buying SCTY Is Neat Idea, but TSLA May Be Asking Too Much of Itself," Piper Jaffray, June 21, 2016, via ThompsonOne, accessed August 2017.

[22]Rod Lache, Mike Levin, Patrick Nolan, and Kirill Babikov, "TSLA/SCTY May Fit, but It's Not a 'No Brainer' to Us," Deutsche Bank Markets Research, June 22, 2016, via ThompsonOne, accessed August 2017.

[23] Colin Langan, Julien Dumoulin-Smith, Paul Zimbardo, and Jerimiah Booream, "A Quick Look at SCTY Board: The Lonely One," UBS, June 24, 2016, via Thompson One, accessed August 2017.

[24] Ryan Brinkman, Samik Chatterjee, and Elad Hillman, "Proposed SolarCity Acquisition Adds to Increasingly Binary and Difficult-to-Value Nature of Tesla Story," J.P. Morgan, June 22, 2016, via ThompsonOne, accessed August 2017.

[25] Tesla Motors / SolarCity Conference Call June 22, 2016. https://forums.tesla.com/forum/forums/tesla-motors-solarcity-conference-call-june-22-2016.

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